

team learning	• mention the advantages and disadvantages of self-directed and team learning • mention the strategies for effective self-directed and team learning • describe the means of better learning and examination performance in MBBS course	• Advantages and disadvantages of self-directed and team learning • Strategies for effective self-directed and team learning • Means of better learning and examination performance in MBBS course		
Medical ethics	At the end of the session students will be able to- • explain the concept of medical ethics • explain the principles, relevance and important issues of medical ethics • state the Hippocratic oath, the International code of medical ethics, the Declaration of Geneva and Important ethical codes of BMDC for a medical doctor	• Concept of medical ethics, principles, purpose/ importance and issues /example of medical ethics • Hippocratic oath • International code of medical ethics • Declaration of Geneva • Ethical codes of BMDC for medical doctors	Interactive Lecture Or Seminar	One and half hour

Integrated Teaching in Phase I

Teachers of all departments of Phase -1 (Anatomy, Physiology & Biochemistry) must be present during these integrated sessions along with the concerned faculties those are mentioned in the column four in the table below. Teachers will be the speakers/facilitators in each session. The students must actively participate in these sessions and have to submit the summary of each session to the concerned teacher/department as their assignments. This assignment will be a part of practical note book in the summative assessment. Students need to get some 'take home message' from every session. Schedule for integrated teaching session will be set at the phase I committee meeting in collaboration with medical education unit (MEU).

Total 36 hour. Each session will be for 3 hour

A) Term-I:

- i. Coronary artery disease
- ii. Chronic obstructive pulmonary disease (COPD)
- iii. Anaemia

B) Term-II:

- iv. Diarrhea
- v. Diabetes Mellitus (DM)
- vi. Jaundice
- vii. Electrolyte imbalance
- viii. Proteinuria

C) Term-III:

- ix. Thyroid disorder
- x. Cerebro-vascular disease (CVD)
- xi. Deafness
- xii. Errors of refraction

Term I

Subject	Learning objective	Core content	Disciplines involved
Coronary artery disease	At the end of the session the student will be able to: - explain the pattern of artery supply of heart - describe the coronary circulation and regulation - explain the appearance & disappearance of cardiac markers with oxygen supply to heart - correlate the knowledge of blood supply of heart obtained in phase I in real life situation	Peculiarity of coronary circulation and its regulation - Balance between supply of blood and demand - Nerve supply of heart and nature of referred pain - ECG changes in ischemic disease Enumerate appearance and disappearance of cardiac markers following ischemic change of coronary artery	Department of Anatomy/ Physiology/ Biochemistry/ Internal Medicine/ Cardiology/ Pathology Time: 3 hours
Chronic obstructive pulmonary disease (COPD)	At the end of the session the student will be able to: - explain the structure and function of respiratory tract - interpret results of spirometry in relation to COPD - differentiate obstructive lung disease from restrictive lung disease - explain the mechanism of acid-base balance, change of pH and PCO₂ in COPD patient - correlate the knowledge of respiratory mechanism in COPD patient obtained in phase I in real life situation	- Different type of epithelium & its specific requirement of that location - Respiratory membrane and factors affecting transport of gases - Spirometry- Pulmonary volume and capacities - Acid-base status in COPD - Change of pH in COPD patient - Mechanism of increased PCO₂ in COPD patient	Department of Anatomy/ Physiology/ Biochemistry/ Internal medicine/ Respiratory medicine Time: 3 hours
Anaemia	At the end of the session the student will be able to: - Define and classify anaemia - Explain role of Hb and RBC in anemia - Interpret red blood cell indices	- Anaemia: Definition, classification - RBC: Erythropoiesis - Haemoglobin: Synthesis, types, functions - Red blood cell indices - Biochemical basis of different types of anaemia	Department of Anatomy/ Physiology/ Biochemistry/ Internal medicine/ Haematology Time: 3 hours

Term II

Subject	Learning objective	Core content	Other discipline involved
Diarrhea	At the end of the session the student will be able to: - explain pattern and function of enteric nerve supply - explain movement of GIT with autonomic effect on it - correlate the consequences of diarrhea	- Enteric nervous system - Gastro-enteric gland distribution - Movements of GIT - Volume disorder occurs in diarrhea - Dehydration in children in diarrhea - Consequence of dehydration	Department of Anatomy/ Physiology/ Biochemistry/ Internal medicine/ Gastro-enterology /

			Paediatrics/ Microbiology/ Pharmacology Time: 3 hours
Diabetes Mellitus (DM)	At the end of the session the student will be able to: • mention the structure and functional relation of Islet of Langerhans • describe structure, mechanism of action, regulation of secretion of insulin • explain pathophysiologic effect of insulin deficiency • explain the metabolism of glucose and changes in DM • develop skill in laboratory diagnosis of DM	• Structure and function of Islet of Langerhans • Islets of Langerhans of pancreas- hormones, functions, mechanism of action, regulation of secretion • Pathophysiology of insulin deficiency • WHO criteria of laboratory diagnosis of DM • Interpretation of OGTT • Metabolic derangement in DM	Department of Anatomy/ Physiology/ Biochemistry/ Internal medicine/ Endocrinology / Time: 3 hours
Jaundice	At the end of the session the student will be able to: • mention structural and functional orientation of hepatocytes • state the steps of bilirubin metabolism • differentiate conjugated & unconjugated bilirubin • define & classify jaundice based on biochemical findings • correlate the knowledge of hepato-biliary system and metabolism obtained in phase I in real life situation	• Role of specific orientation of hepatocyte • Relation of intrahepatic biliary tree and hepatocyte • Steps of bilirubin metabolism • Conjugated & unconjugated bilirubin • Jaundice based on biochemical findings	Department of Anatomy/ Physiology/ Biochemistry/ Internal medicine/ Haematology Time: 3 hours
Electrolyte imbalance	At the end of the session the student will be able to: • explain homeostatic functions of kidney for the regulation of electrolytes • correlate normal electrolyte level, its deviation & consequences of deviation	• Homeostatic function of kidney • Regulation of electrolytes by hormones acting on kidney • Laboratory result of electrolyte profile • Consequences of different types of electrolytes imbalance	Department of Physiology/ Biochemistry/ Internal medicine/ Nephrology / Anesthesiology Time: 3 hours
Proteinuria	At the end of the session the student will be able to: • describe glomerular membrane, GFR, effective filtration pressure • correlate the structure and function of filtration membrane Explain consequences of proteinuria. • explain consequences of proteinuria.	• GFR: definition, determinants and control • Normal reabsorption process in kidney • Proteinuria: Detection, pathophysiology of developing proteinuria, important causes.	Department of Anatomy/ Physiology/ Biochemistry/ Internal medicine/ Nephrology / Paediatrics Time: 3 hours

Term III

Subject	Learning objective	Core content	Other discipline involved
Thyroid disorder	At the end of the session the student will be able to: • mention structure of thyroid gland • describe biosynthesis, storage, release, transport, mechanism of action, function and regulation of secretion of thyroid hormone • explain the importance iodine in thyroid hormone synthesis • interpret the thyroid function test	• Structure of thyroid gland • Thyroid hormone biosynthesis, storage, release, transport, mechanism of action, function and regulation of secretion of thyroid hormone • Thyroid disorders: hypo and hyperthyroidism, cretinism, myxoedema and goitre • Importance of iodine in thyroid hormone synthesis • Thyroid function tests with their interpretation	Department of Anatomy/ Physiology/ Biochemistry/ Internal medicine/ Endocrinology Time: 3 hours
Cerebro-vascular disease (CVD)	At the end of the session the student will be able to: • explain the blood supply of CNS • explain the pattern and functioning of blood brain barrier • explain effect of UMN & LMN lesion • interpret deep & superficial reflexes • correlate the knowledge of blood supply of CNS obtained in phase I in real life situation	• Peculiarity of artery supply of CNS • Blood brain barrier • Ascending and descending tracts: name and functions. • UMN & LMN: definition, effect of lesion • Role of dyslipidemia in developing CVD.	Department of Anatomy/ Physiology/ Biochemistry/ Internal medicine/ Neurology Time: 3 hours
Deafness	At the end of the session the student will be able to: • explain the role of different organs of hearing • explain sound wave transmission, excitation of auditory receptors, auditory pathway • interpret the result of Rinne test & Weber test.	• Role of different parts/organs in hearing • Hearing: receptor, mechanism of sound wave transmission, auditory pathway.	Department of Anatomy/ Physiology/ Biochemistry/ Otolaryngology Time: 3 hours
Errors of refraction	At the end of the session the student will be able to: • Summarise the structure of eye ball, refractive media, refractive index, diaptor, refractive power of cornea & lens, • types, causes of errors of refraction and their correction	• Structure of eye ball • Vision: image formation in the eye, visual pathway, common errors of refraction.	Department of Anatomy/ Physiology/ Biochemistry/ Ophthalmology Time: 3 hours